Finite-element formulation of a dental drill cavity with recurrent-prime path perturbations

A finite, reproducible 2-D material-field surrogate with a bounded continued-fraction path law

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Prepared from the repository's deterministic drill-path simulation; not affiliated with a journal.

Abstract. Dental drilling requires a cavity trajectory that is geometrically bounded, reproducible and interpretable against a material model. We formulate a finite-element problem on a cavity-excised domain and instantiate its geometry using a deterministic 2-D material-field surrogate. The centreline is perturbed by a finite set of eight prime harmonics, whose phases are fixed by convergents of the continued fractions of square roots. The terminal taper makes both entry and terminal lateral offsets exactly zero. For the repository default (4.0 mm planned depth, 0.075 mm amplitude), the 38 x 52-cell field reports a maximum lateral correction of 6.92 micrometres, peak non-drill tissue stress of 1.183 MPa, peak tool stress of 83.65 MPa and peak cutting force of 6.42 N. These are deterministic surrogate outputs, not measurements, clinical predictions or evidence of safety. The manuscript provides the finite equations, boundary definitions and parameter values needed to reproduce the calculation.

Scope statement. This document describes research software only. It does not use patient data, is not connected to hardware, and must not be used to select a drilling trajectory or make clinical decisions. The present implementation is a deterministic 2-D material-field surrogate, rather than a validated patient-specific three-dimensional finite-element analysis.

1. Problem definition

Let the rectangular computational domain be $\Omega = [-4.5, 4.5]\ \text{mm} \times [-0.8, 5.2]\ \text{mm}$, with depth increasing from the entrance surface toward the cavity floor. At normalized progress $u \in [0, 1]$, the planned axial coordinate is $z(u) = Du$ with $D = 4.0$ mm. A 0.52 mm-wide moving drill envelope is removed from the hard-tissue proxy. The remaining finite domain is divided into soft-tissue boundary, hard-tissue proxy and pulp-protection boundary labels. Figure 1 gives the coordinate convention.

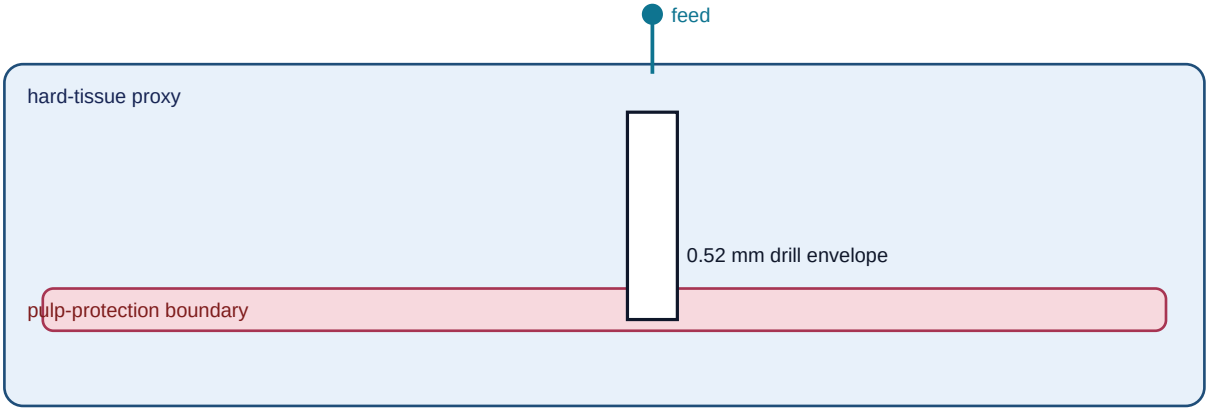

Figure 1 | Finite cavity domain and moving envelope. *The blue region is the hard-tissue proxy; pink marks the protected boundary. The central white channel is the 0.52 mm drill envelope. This is a schematic of the deterministic material map, not anatomical geometry.*

2. Finite-element formulation

For an elastic interpretation, the cavity-excised domain Ω_h is partitioned into a finite mesh T_h of elements e. The displacement approximation uses finite nodal basis functions N_a. Essential boundary conditions are imposed on Γ_u; a distributed cutting load acts on the cavity boundary Γ_c. This standard weak form makes explicit what a full 3-D implementation would solve; the current code evaluates the 2-D deterministic stress proxy described in Section 4 instead.

$$V_h = \{ v_h = \sum_{a=1}^{N} N_a v_a : v_h|_{\Gamma_u} = 0 \}, \quad \Omega_h = \Omega \setminus B_r(c(u))$$
$$\sum_{e \in T_h} \int_{\Omega_e} \varepsilon(v_h)^T C_e \varepsilon(u_h)\, d\Omega = \int_{\Gamma_c} v_h^T t_c\, d\Gamma \tag{1}$$

Equation 1 | Finite weak form on the cavity-excised domain. B_r is the drill envelope, $\boldsymbol{C}_e$ is an element constitutive matrix, and $\boldsymbol{t}_c$ is a boundary traction.

$$KU = F, \quad K = \sum_{e \in T_h} K_e', \quad K_e = \int_{\Omega_e} B_e^T C_e B_e \, d\Omega$$
$$\sigma_{vM} = \sqrt{ \frac{ [\sigma_{xx} - \sigma_{yy}]^2 + \sigma_{yy}^2 + \sigma_{xx}^2 + 6\tau_{xy}^2 }{2} } \tag{2}$$

Equation 2 | Assembly of the finite linear system and plane-stress von Mises post-processing expression.

3. Recurrent-prime centreline

The centreline is a bounded, finite trigonometric sum. The prime set is P = {2, 3, 5, 7, 11, 13, 17, 19}. For each p_k, a six-term continued fraction for $\sqrt{p_k}$ produces a convergent n_k/d_k; the phase c_k is the fractional component of that convergent. Weights are $w_k = \ln(p_k)/\sqrt{p_k}$. The taper $u(1-u)^2$ forces x(0) = x(1) = 0 independently of the phases, so the perturbation does not change the entry or terminal coordinate.

$$c_k = (n_k \bmod d_k) / d_k, \quad w_k = \ln(p_k) / \sqrt{p_k}, \quad z(u) = Du$$
$$x(u) = A\, u(1-u)^2 \left\{ \frac{\sum_{k=1}^{8} w_k \sin[2\pi p_k u + 2\pi c_k]}{\sum_{k=1}^{8} w_k} \right\}, \quad A = 0.075 \text{ mm}, \quad D = 4.0 \text{ mm} \tag{3}$$

Equation 3 | Bounded recurrent-prime path law used by the simulation. All sums, phases and parameters are finite.

p_k	six-term convergent	phase c_k
2	99/70	0.4142857
3	26/15	0.7333333
5	2889/1292	0.2360681
7	45/17	0.6470588
11	3970/1197	0.3166249
13	119/33	0.6060606
17	143649/34840	0.1231056
19	170/39	0.3589744

Table 1 | Prime phases fixed by square-root continued-fraction convergents. *These values are generated deterministically by the repository simulator.*

4. Implemented material-field surrogate

The current implementation samples 52 lateral locations and 38 depth locations. Each cell receives a categorical material label and a scalar stress proxy. The drill envelope receives zero tissue stress. Outside this envelope, two compact kernels concentrate the field near the advancing tip and entrance interface. This is a 2-D deterministic field used for visualization and sensitivity checks; it is not a substitute for material calibration, contact mechanics, thermal modelling, crack propagation or a converged 3-D mesh study.

$$I_{drill}(x,z;u) = 1\{ 0 \le z \le D \text{ and } |x - x(u)| \le 0.26 \text{ mm} \},$$
$$s(x,z;u) = [1 - I_{drill}] \left[0.08 + 1.15 \exp(-d_{tip}^2) + 0.34 \exp(-d_{entry}^2)\right] \text{MPa},$$
$$d_{tip}^2 = \left(\frac{|x - x(u)|}{0.48}\right)^2 + \left(\frac{z - D}{0.72}\right)^2, \quad d_{entry}^2 = \left(\frac{|x - x(u)|}{0.88}\right)^2 + \left(\frac{z - 0.12}{0.55}\right)^2 \tag{4}$$

Equation 4 | Finite 2-D stress-proxy field evaluated on the 38 x 52 grid. The dimensions in the kernels are millimetres.

5. Default simulation output

One deterministic run was generated with the source defaults. Because the model contains no random sampling, repeated runs with the same code and parameters yield the same values. The peak tissue result excludes the drill envelope, consistent with Equation 4. Reported decimal values are implementation outputs and should be interpreted only within the surrogate's stated scope.

Quantity	Default output	Interpretation
Planned depth	4.0 mm	Axial path extent
Lateral amplitude A	0.075 mm	Bound before terminal taper
Grid	38 x 52 cells	Deterministic 2-D material field
Maximum correction	6.92 micrometres	max |x(u)| along sampled path
Entry / terminal offset	0.00 / 0.00 micrometres	Guaranteed by the taper
Peak tissue stress proxy	1.183 MPa	Maximum outside drill cells
Peak tool stress proxy	83.65 MPa	Trajectory trace maximum
Peak cutting-force proxy	6.42 N	Trajectory trace maximum

Table 2* | *Reproducible default outputs. *Values were produced by* `RecurrentPrimeDrillPathEngine().simulate()`.

6. Verification and limitations

The following algebraic checks are directly verifiable from the model: (i) the finite prime list has cardinality eight; (ii) every phase is a rational number obtained from a finite convergent; (iii) the normalization denominator is positive because $p_k > 1$; and (iv) x(0) = x(1) = 0. The last identity follows from the taper and does not depend on harmonic cancellation. These checks establish reproducibility of the path definition, not biomechanical validity.

$$x(0) = A\cdot 0\cdot(1-0)^2 R(0) = 0, \quad x(1) = A\cdot 1\cdot(1-1)^2 R(1) = 0,$$
$$R(u) = \left\{ \sum_{k=1}^{8} w_k \sin(2\pi p_k u + 2\pi c_k) \right\} \Big/ \sum_{k=1}^{8} w_k, \quad \sum_{k=1}^{8} w_k > 0 \tag{5}$$

Equation 5 | Endpoint invariance of the recurrent-prime perturbation.

The results should not be extrapolated to tooth-specific morphology, enamel, dentin, cementum, pulp, periodontal ligament, irrigation, heat generation, burr wear, operator control, robotic positioning error or fracture risk. A clinically relevant study would require image-derived geometry, material calibration, mesh-convergence analysis, loading and thermal boundary conditions, uncertainty quantification, independent verification and appropriate regulatory validation.

7. Reproducibility

The generator is self-contained apart from ReportLab and the local `drill_path_planner.py` simulation engine. Run `python3 dental_robot_surgery/generate_fea_cavity_preprint.py` from the repository root to regenerate this document. The source specifies the default depth, amplitude, finite prime set, continued-fraction length, grid and equations; no external data are required.

References

1. Hughes, T. J. R. *The Finite Element Method: Linear Static and Dynamic Finite Element Analysis*. Dover (2000).
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4. RecurrentPrimeDrillPathEngine. Local deterministic research simulation in `dental_robot_surgery/drill_path_planner.py` (accessed 24 October 2026).